Cryptography and Network Security (BT14503) :
Classical Encryption Techniques

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Outline

- Symmetric Cipher Model
  - Cryptography
  - Cryptanalysis and Brute-Force Attack
- Substitution Techniques
  - Caesar Cipher
  - Monoalphabetic Ciphers
  - Polyalphabetic Ciphers
  - Playfair Cipher
  - Hill Cipher
  - Vigenère Ciphers
  - One-Time Pad
  - Rail Fence Cipher
  - Row Transposition Cipher
- Steganography

Basic Terms

- An original message is known as the plaintext,
  - while the coded message is called the ciphertext.
- The process of converting from plaintext to ciphertext is known as enciphering or encryption;
  - restoring the plaintext from the ciphertext is deciphering or decryption.
- Symmetric encryption,
  - also referred as conventional encryption or single-key encryption

Symmetric Cipher Model

- **Plaintext:**
  - This is the original intelligible message or data that is fed into the algorithm as input.
- **Encryption algorithm:**
  - The encryption algorithm performs various substitutions and transformations on the plaintext.
- **Secret key:**
  - The secret key is also input to the encryption algorithm.
  - The key is a value independent of the plaintext and of the algorithm.
  - The algorithm will produce a different output depending on the specific key being used at the time.
  - The exact substitutions and transformations performed by the algorithm depend on the key.

Symmetric Cipher Model

- **Ciphertext:**
  - This is the scrambled message produced as output.
  - It depends on the plaintext and the secret key.
    - For a given message, two different keys will produce two different ciphertexts. The ciphertext is an apparently random stream of data and, as it stands, is unintelligible.
- **Decryption algorithm:**
  - This is essentially the encryption algorithm run in reverse.
  - It takes the ciphertext and the secret key and produces the original plaintext.

Symmetric Cipher Model..

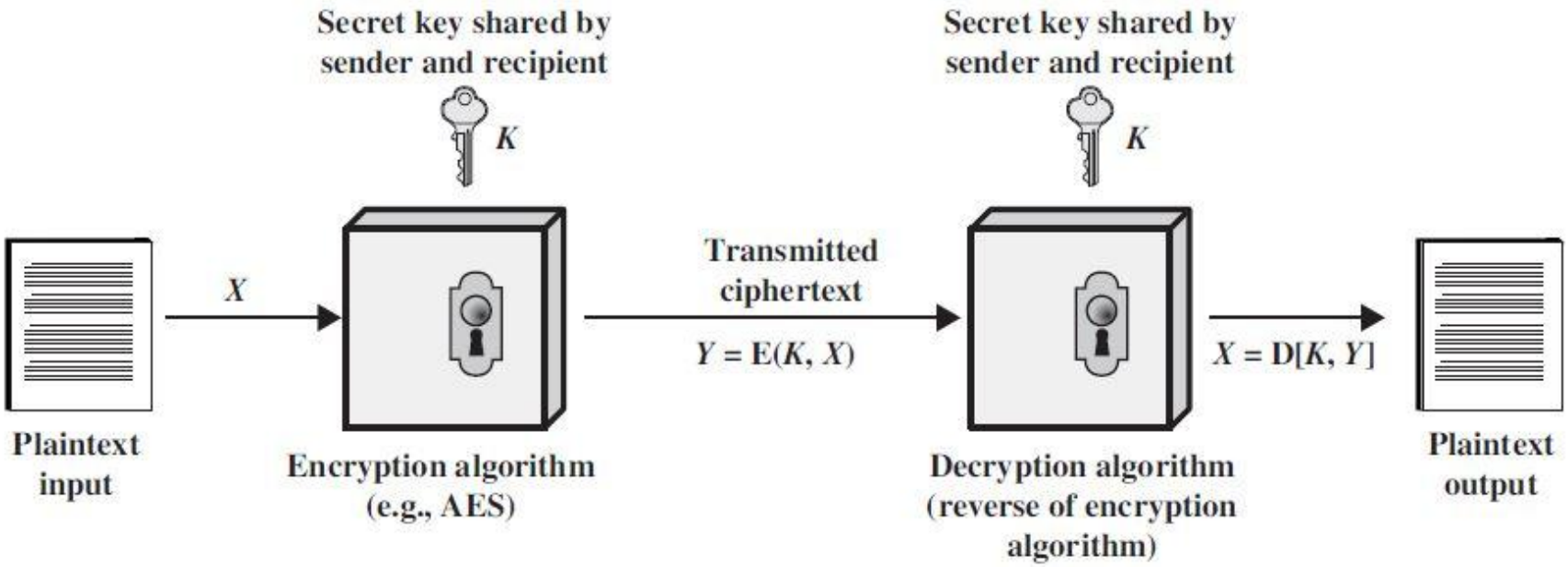


Fig 1: Simplified Model of Symmetric Encryption

Symmetric Cipher Model..

- There are two requirements for secure use of conventional encryption:
  - A strong encryption algorithm
  - Sender and receiver must have obtained copies of the secret key in a secure fashion and must keep the key secure

Symmetric Cipher Model..

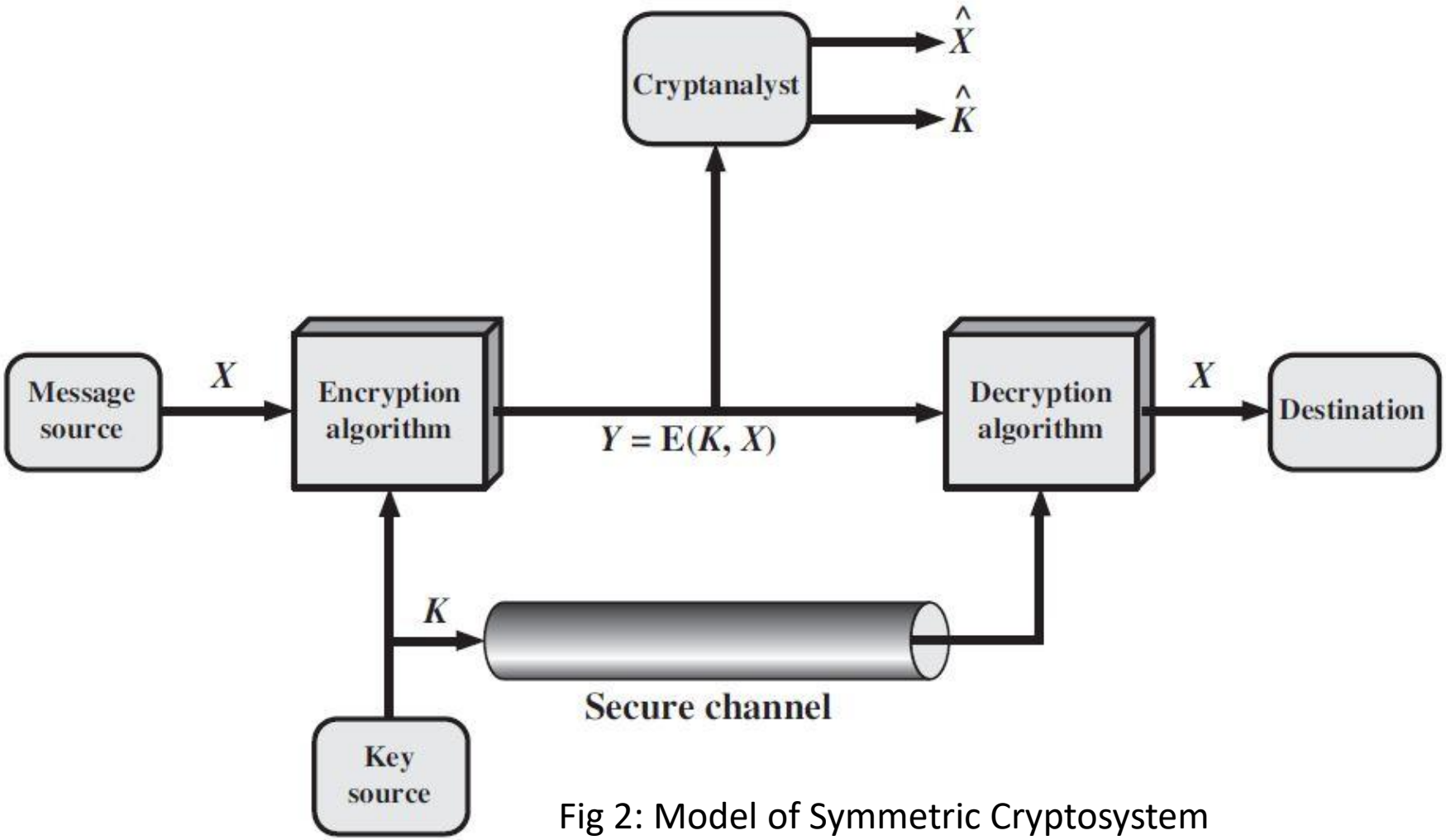


Fig 2: Model of Symmetric Cryptosystem

Symmetric Cipher Model..

- Plaintext $X = [X_1, X_2, X_3 \ \ldots . X_M]$
- Key $K = [K_1, K_2, K_3 \ \ldots . K_J]$
- Ciphertext $Y = [Y_1, Y_2, Y_3 \ \ldots . Y_N]$
- Mathematically have:

$$Y = E_K(X)$$
$$X = D_K(Y)$$

- Opponent knows the encryption (E) and decryption (D) algorithms.
  - Observes Y but no access to X or K
  - Attempt to recover X, or K, or both X and K

Cryptography

- Characterize cryptographic systems by:

1. **The type of operations used for transforming plaintext to ciphertext.**
   - All encryption algorithms are based on two general principles:
     - substitution, in which each element in the plaintext (bit, letter, group of bits or letters) is mapped into another element,
     - transposition, in which elements in the plaintext are rearranged.
   - Most systems, referred to as *product systems*, involve multiple stages of substitutions and transpositions.

Cryptography

- Characterize cryptographic systems by:

2. The number of keys used.

  - If both sender and receiver use the same key, the system is referred to as symmetric, single-key, secret-key, or conventional encryption.
  - If the sender and receiver use different keys, the system is referred to as asymmetric, two-key, or public-key encryption.

Cryptography

- Characterize cryptographic systems by:

3. The way in which the plaintext is processed.

- A *block cipher* processes the input one block of elements at a time, producing an output block for each input block.
- A *stream cipher* processes the input elements continuously, producing output one element at a time, as it goes along.

Cryptanalysis

- The process of attempting to discover the plaintext or key is known as **cryptanalysis**
- The strategy used by the cryptanalyst depends on
  - nature of the encryption scheme and
  - information available to the cryptanalyst

Cryptanalysis Attack

- Ciphertext-Only Attack
  - Adversary has access to only some ciphertext
    - Attempts to find the corresponding key and the plaintext.
  - most probable one because Eve needs only the ciphertext for this attack

Cryptanalysis Attack..

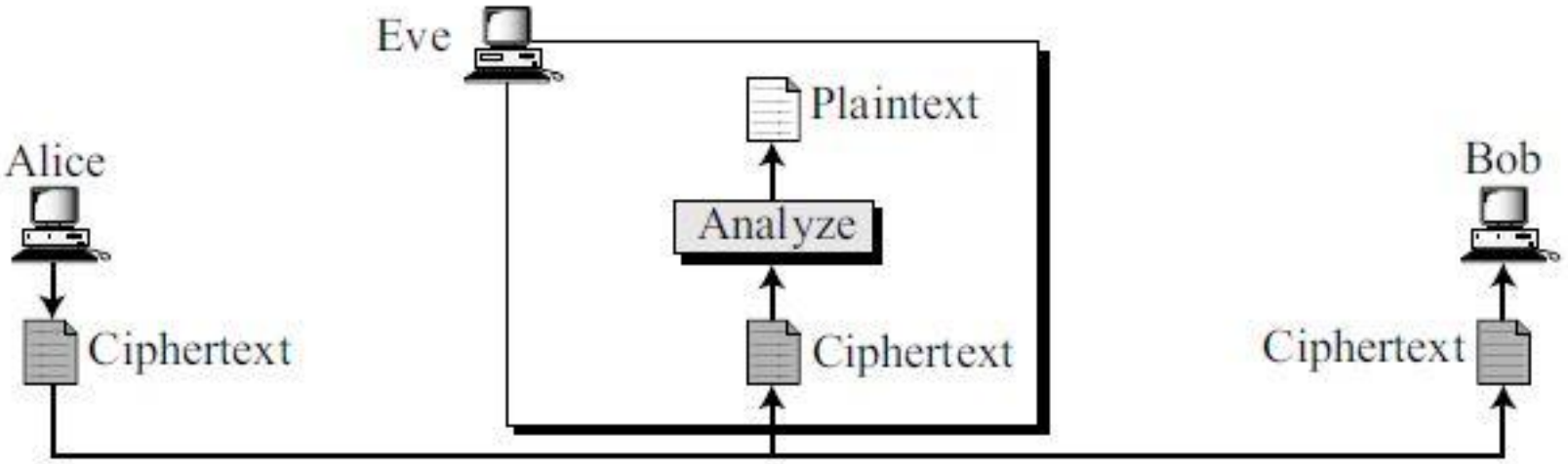


Fig 3: Ciphertext-Only Attack

Cryptanalysis Attack..

- Known-Plaintext Attack
  - Adversary has access to some plaintext/ciphertext pairs
    - in addition to the intercepted ciphertext that to break
  - The plaintext/ciphertext pairs have been collected earlier.
    - For example, Alice has sent a secret message to Bob, but she has later made the contents of the message public.
    - Eve has kept both the ciphertext and the plaintext to use them to break the next secret message from Alice to Bob, assuming that Alice has not changed her key.
    - Eve uses the relationship between the previous pair to analyze the current ciphertext
  - easier to implement
    - because adversary has more information to use for analysis
  - less likely to happen
    - because sender may have changed her key or may have not disclosed the contents of any previous messages

Cryptanalysis Attack

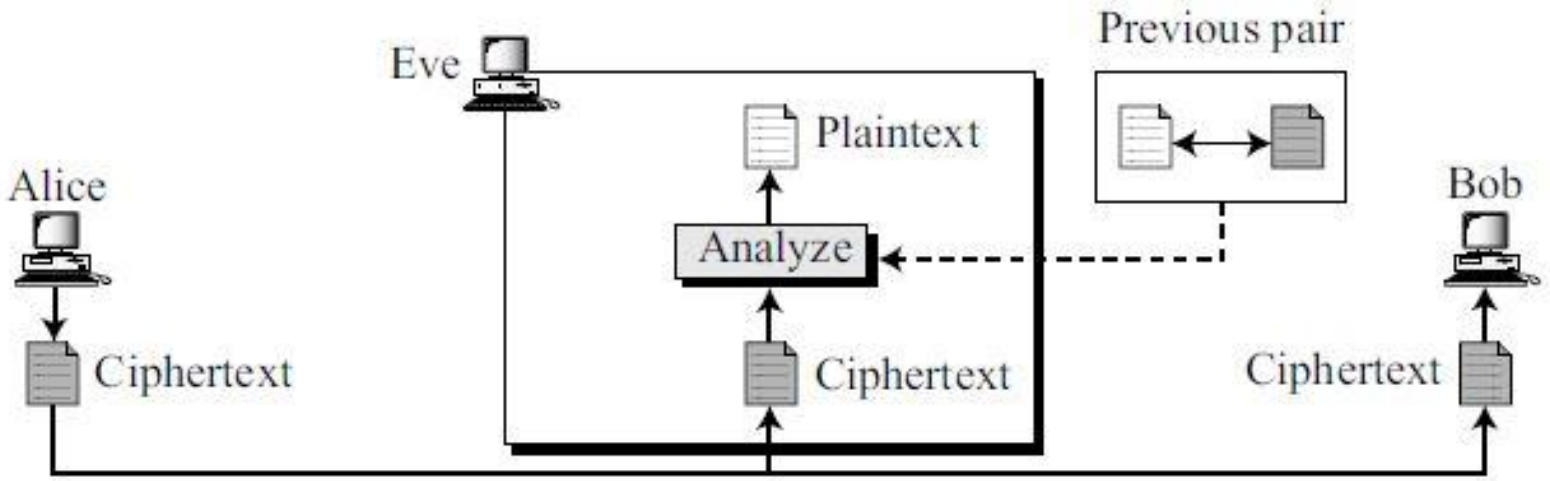


Fig 4: Known-Plaintext Attack

Cryptanalysis Attack..

- Chosen-Plaintext Attack
  - similar to the known-plaintext attack,
    - but the plaintext/ciphertext pairs have been chosen by the attacker herself
  - Adversary has access to sender's computer.
    - He/she can choose some plaintext and intercept the created ciphertext.
    - Adversary has no access to key
  - Attack is easier to implement
    - but less likely to happen

Cryptanalysis Attack..

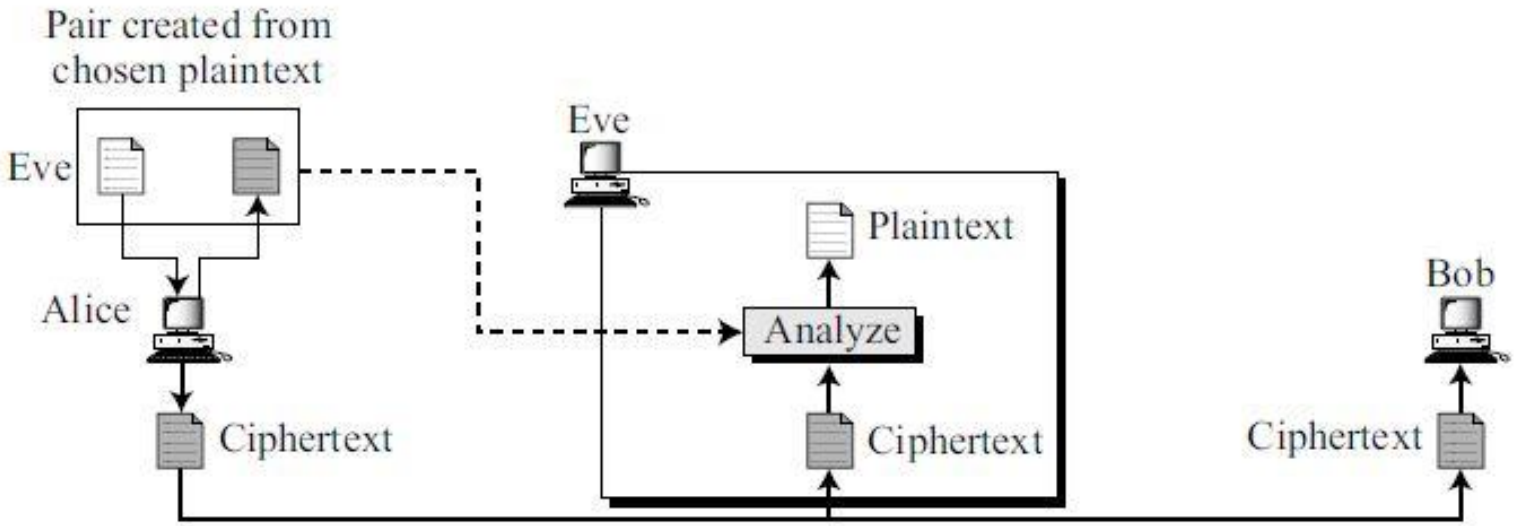


Fig 5: Chosen-Plaintext Attack

Cryptanalysis Attack..

- Chosen-Ciphertext Attack
  - similar to the chosen-plaintext attack,
    - except that Adversary chooses some ciphertext
      - and decrypts it to form a ciphertext/plaintext pair
    - This can happen if Adversary has access to receiver's computer

Cryptanalysis Attack..

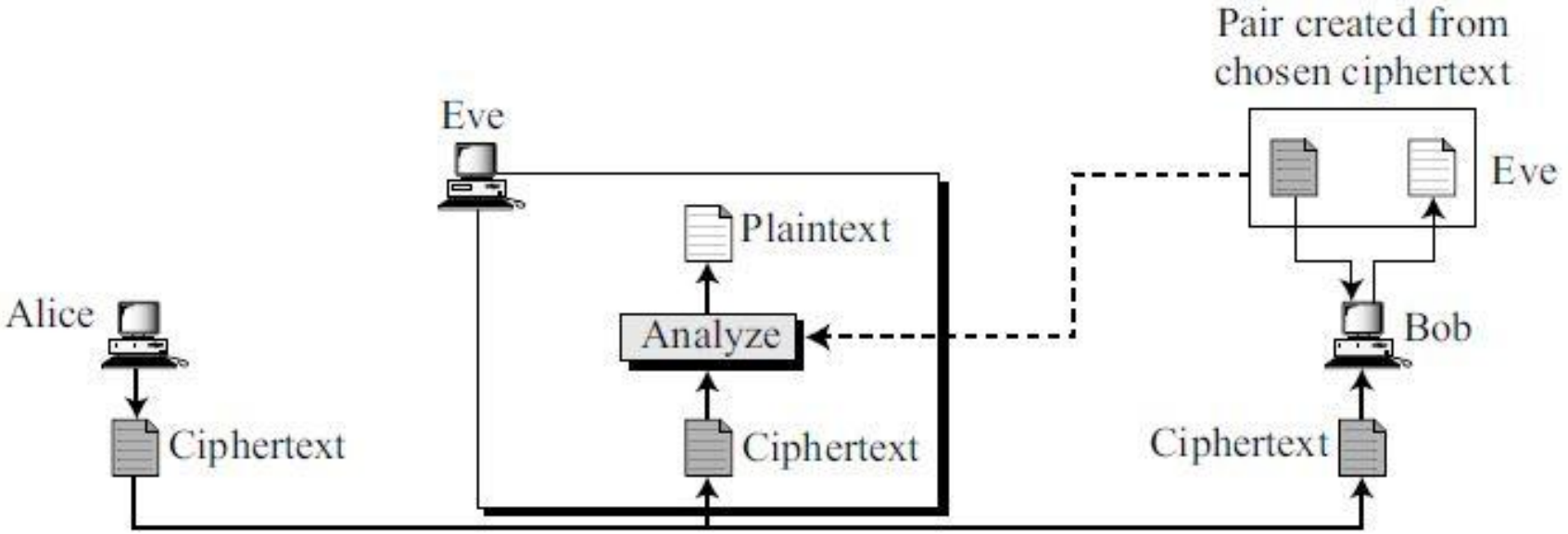


Fig 6: Chosen-Ciphertext Attack

Cryptanalysis Attack..

- Brute-force Attack
  - Also called as exhaustive-key-search method
  - Adversary knows
    - algorithm and
    - key domain (list of all possible keys)
  - Adversary uses intercepted cipher,
    - and decrypts the ciphertext with every possible key until the plaintext makes sense
  - To prevent this type of attack, the number of possible keys must be very large

Cryptanalysis Attack..

- An encryption scheme is **computationally secure** if the ciphertext generated
- by the scheme meets one or both of the following criteria:
  - The cost of breaking the cipher exceeds the value of the encrypted information.
  - The time required to break the cipher exceeds the useful lifetime of the information.

Cryptanalysis Attack..

- Statistical Attack
  - cryptanalyst can benefit from some inherent characteristics of the plaintext language
    - For example, we know that the letter E is the most frequently used letter in English text.
      - The cryptanalyst finds the mostly-used character in the ciphertext and assumes that the corresponding plaintext character is E.
      - After finding a few pairs, the analyst can find the key and use it to decrypt the message.

Substitution Ciphers

- Is one in which the letters of plaintext are replaced by other letters or by numbers or symbols
- If the plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns

Caesar Cipher

- Simplest and earliest known use of a substitution cipher
- Also referred as additive cipher or shift cipher
- Used by Julius Caesar
- Involves replacing each letter of the alphabet with the letter standing three places further down the alphabet
- Alphabet is wrapped around so that the letter following Z is A

plain: `meet me after the party`

cipher: `PHHW PH DIWHU WKH SDUWB`

Caesar Cipher Algorithm

- Can define transformation as:

```
a b c d e f g h i j k l m n o p q r s t u v w x y z
D E F G H I J K L M N O P Q R S T U V W X Y Z A B C
```

- Mathematically give each letter a number

```
a b c d e f g h i j  k  l  m  n  o  p  q  r  s  t  u  v  w  x  y  z
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
```

- Algorithm can be expressed as:

$$c = E(3, p) = (p + 3) \bmod (26)$$

- A shift may be of any amount, so that the general Caesar algorithm is:

$$C = E(k, p) = (p + k) \bmod 26$$

- Where k takes on a value in the range 1 to 25; the decryption algorithm is simply:

$$p = D(k, C) = (C - k) \bmod 26$$

Fig 7:Brute-Force Cryptanalysis of Caesar Cipher

KEY	PHHW PH DIWHU WKH WRJD SDUWB
1	oggv og chvgt vjg vqic rctva
2	nffu nf bgufs uif uphb qbsuz
3	meet me after the toga party
4	ldds ld zesdq sgd snfz ozqsx
5	kccr kc ydrcp rfc rmey nyprw
6	jbbq jb xcqbo qeb qldx mxoqv
7	iaap ia wbpan pda pkcw lwnpu
8	hzzo hz vaozm ocz ojbv kvmot
9	gyyn gy uznyl nby niau julns
10	fxxm fx tymxk max mhzt itkmr
11	ewwl ew sxlwj lzw lgys hsjlq
12	dvvk dv rwkvi kyv kfxr grikp
13	cuuj cu qvjuh jxu jewq fqhjo
14	btti bt puitg iwt idvp epgin
15	assh as othsf hvs hcuo dofhm
16	zrrg zr nsgre gur gbtn cnegl
17	yqqf yq mrfqd ftq fasm bmdfk
18	xppe xp lqepc esp ezrl alcej
19	wood wo kpdob dro dyqk zkbdi
20	vnnc vn jocna cqn cxpj yjach
21	ummb um inbmz bpm bwoi xizbg
22	tlla tl hmaly aol avnh whyaf
23	skkz sk glzkx znk zumg vgxze
24	rjjy rj fkyjw ymj ytlf ufwyd
25	qiix qi ejxiv xli xske tevxc

Monoalphabetic Cipher

- With only 25 possible keys, the Caesar cipher is far from secure.
- A dramatic increase in the key space can be achieved by allowing an arbitrary substitution
- Permutation
  - Of a finite set of elements *S* is an ordered sequence of all the elements of *S* , with each element appearing exactly once
  - For example, if S = {a, b, c}, there are six permutations of S : abc, acb, bac, bca, cab, cba

Monoalphabetic Cipher..

- In general, there are $n!$ permutations of a set of elements,
  - because the first element can be chosen in one of n ways,
  - the second in $(n-1)$ ways,
  - the third in $(n-2)$ ways, and so on.
- If the "cipher" line can be
  - any permutation of the 26 alphabetic characters,
    - then there are 26! possible keys
- Such an approach is referred to as a **monoalphabetic substitution cipher**,
  - because a single cipher alphabet (mapping from plain alphabet to cipher alphabet) is used per message.

Monoalphabetic Cipher..

- Attack is still possible.
- If the cryptanalyst knows the nature of the plaintext,
  - then the analyst can exploit the regularities of the language.
  - For example, consider ciphertext

```
UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAI
ZVUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSXEPYE
POPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ
```

Monoalphabetic Cipher..

- As a first step, the relative frequency of the letters can be determined and compared to a standard frequency distribution for English.
  - If the message were long enough, this technique alone might be sufficient, but because this is a relatively short message, we cannot expect an exact match.

P 13.33	H 5.83	F 3.33	B 1.67	C 0.00
Z 11.67	D 5.00	W 3.33	G 1.67	K 0.00
S 8.33	E 5.00	Q 2.50	Y 1.67	L 0.00
U 8.33	V 4.17	T 2.50	I 0.83	N 0.00
O 7.50	X 4.17	A 1.67	J 0.83	R 0.00
M 6.67				

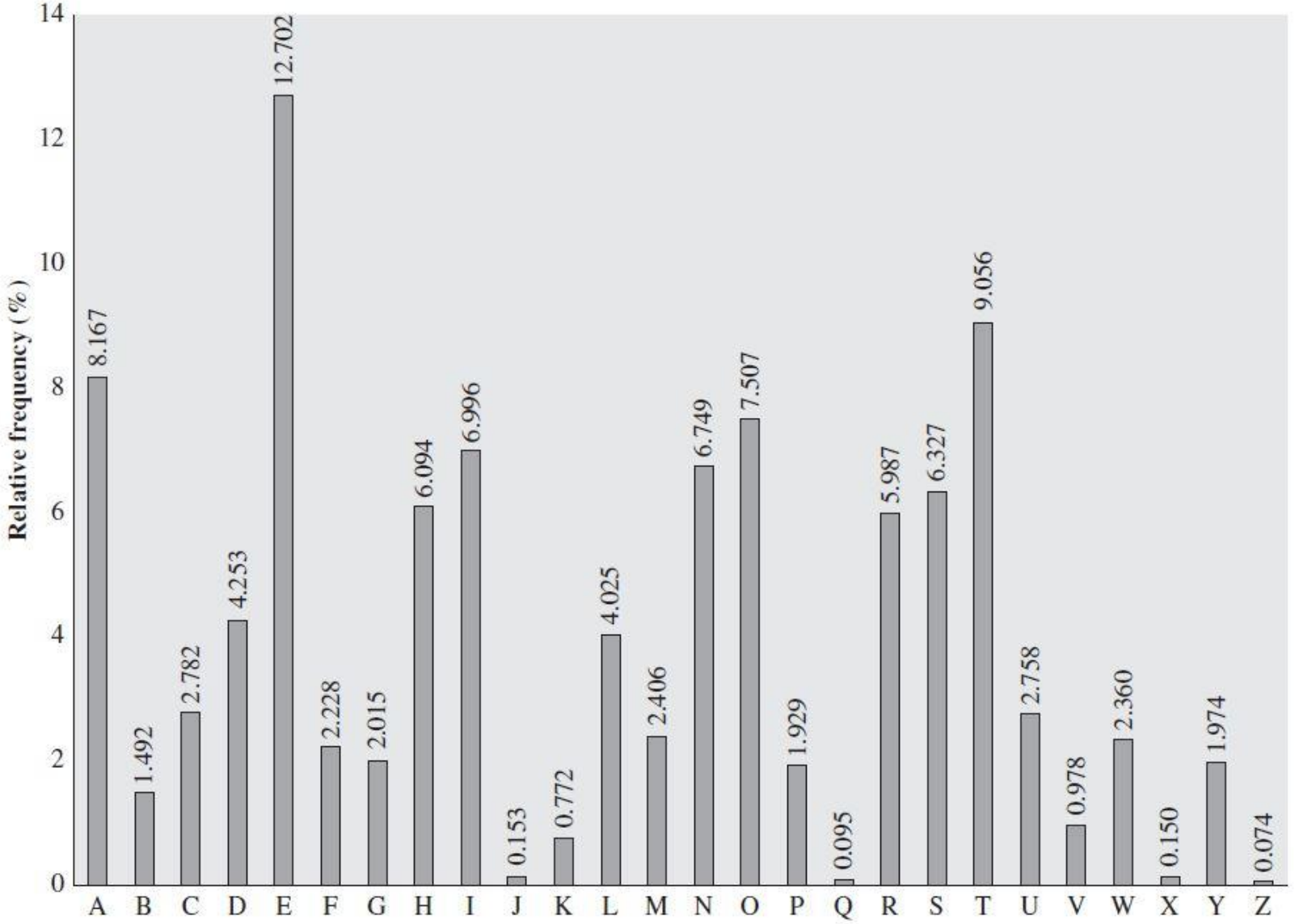


Fig 8: Relative Frequency of Letters in English Text

Monoalphabetic Cipher..

- guess P & Z are e and t
- guess ZW is th and hence ZWP is “the”
- proceeding with trial and error finally get:

  it was disclosed yesterday that several informal but direct contacts have been made with political representatives of the viet cong in mosco

Monoalphabetic Cipher..

- Easy to break because they reflect the frequency data of the original alphabet
  - Digram
    - Two-letter combination
    - Most common is *th*
  - Trigram
    - Three-letter combination
    - Most frequent is *the*
- Countermeasure is to provide multiple substitutes (homophones) for a single letter

Monoalphabetic Cipher..

Letter	*Frequency*	*Letter*	*Frequency*	*Letter*	*Frequency*	*Letter*	*Frequency*
E	12.7	H	6.1	W	2.3	K	0.08
T	9.1	R	6.0	F	2.2	J	0.02
A	8.2	D	4.3	G	2.0	Q	0.01
O	7.5	L	4.0	Y	2.0	X	0.01
I	7.0	C	2.8	P	1.9	Z	0.01
N	6.7	U	2.8	B	1.5		
S	6.3	M	2.4	V	1.0		

Table 1: Frequency of occurrence of letters in an English text

Monoalphabetic Cipher..

Digram	TH, HE, IN, ER, AN, RE, ED, ON, ES, ST, EN, AT, TO, NT, HA, ND, OU, EA, NG, AS, OR, TI, IS, ET, IT, AR, TE, SE, HI, OF
Trigram	THE, ING, AND, HER, ERE, ENT, THA, NTH, WAS, ETH, FOR, DTH

Table 2: Grouping of digrams and trigrams based on their frequency in English

Polyalphabetic Substitution

- In polyalphabetic substitution, each occurrence of a character may have a different substitute.
- The relationship between a character in the plaintext to a character in the ciphertext is one-to-many.
  - For example, “a” could be enciphered as “D” in the beginning of the text, but as “N” at the middle.
- Polyalphabetic ciphers have the advantage of hiding the letter frequency of the underlying language
  - Examples: Playfair Cipher, Hill Cipher, Vigenere Cipher

Playfair Cipher

- A brute-force attack on a Playfair cipher is very difficult. The size of the key domain is $25!$
  - Additionally, encipherment hides the single-letter frequency of the characters.
  - However, the frequencies of diagrams are preserved to some extent.
    - so cryptanalyst can use digram frequency to find the key.

Hill Cipher

- Attacker can perform Known-Plaintext attack
  - in which attacker knows plaintext/ciphertext pairs for at least m blocks
- Attacker creates two $m \times m$ matrices
  - X (plaintext) and C (ciphertext) in which the corresponding rows represent the corresponding known plaintext/ciphertext pairs
  - Because $C = XK$, Attacker uses relation $K = CX^{-1}$ to find key if X is invertible
    - If X is not invertible, Attacker needs different set of m plaintext/ciphertext pairs

Vigenère Ciphers

- In this scheme the set of related monoalphabetic substitution rules consists of the 26 Caesar ciphers with shifts of 0 through 25
- The key stream is a repetition of an initial secret key stream of length m, where we have $1 \leq m \leq 26$
- The mapping of plaintext $\mathrm{P} = P_1 P_2 P_3$ … to ciphertext $\mathrm{C} = C_1 C_2 C_3$ is defined on individual characters by Encryption $C_i = P_i + k_i$ and Decryption $P_i = C_i - k_i$

Vigenère Ciphers

- Example:
- Plaintext- `She is listening`
- Keyword- `Pascal`

Plain Text	s	h	e	i	s	l	i	s	t	e	n	i	n	g
P's Value	18	7	4	8	18	11	8	18	19	4	13	8	13	6
Key	p	a	s	c	a	l	p	a	s	c	a	l	p	a
K's Value	15	0	18	2	0	11	15	0	18	2	0	11	15	0
C's Value	7	7	22	10	18	22	23	18	11	6	13	19	2	6
Ciphertext	h	h	w	k	s	w	x	s	l	g	n	t	c	g

Vigenère Ciphers – Vigenère Table

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	v	v	w	x	y	z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Vigenère Ciphers

- The first-row shows
  - plaintext character to be encrypted.
- The first column
  - shows the characters to be used by the key.
- The rest of the tableau shows the ciphertext characters.
  - To find the ciphertext for the plaintext “she is listening” using the word
  - “PASCAL” as the key, “She is listening” as plaintext
    - we can find “s” in the first row, “P” in the first column, the cross section is the ciphertext character “H”.
    - We can find “h” in the first row and “A” in the second column, the cross section is the ciphertext character “H”.

Vigenère Ciphers

- Even this scheme is vulnerable to cryptanalysis
  - Because the key and the plaintext share the same frequency distribution of letters, a statistical technique can be applied

Vigenère Ciphers – One Time Pad

- Use random key
  - as long as plaintext so that key is not repeated
- Key should be used to encrypt and decrypt plaintext
  - and then discarded
- This is called as one-time pad
  - as it produces random output which has no statistical relationship to plaintext

Vigenère Ciphers – One Time Pad..

- There are two fundamental difficulties:
  - Practical problem in making large quantities of random keys of same size as plaintext
  - Problem of securing key distribution and protection
    - For every message, key of equal length is needed by both sender and receiver

Steganography: Information Hiding

- Steganography conceal the message itself by covering it with something else.
- Steganography hides the existence of a message by transmitting information through various carriers. Its goal is to prevent the detection of a secret message.
- **Steganography** is the art and science of writing hidden messages in such a way that no one, apart from the sender and intended recipient, suspects the existence of the message

Steganography – Classical Examples

- Tap Code

	1	2	3	4	5
1	**A** . .	**B** . ..	**C, K** . …	**D** . ….	**E** . …..
2	**F** .. .	**G**	**H** .. …	**I** .. ….	**J** .. …..
3	**L** … .	**M** … ..	**N** … …	**O** … ….	**P** … …..
4	**Q** … .	**R** …. ..	**S** …. …	**T** …. ….	**U** …. …..
5	**V** ….. .	**W** ….. ..	**X** ….. …	**Y** ….. ….	**Z** ….. …..

- **….. .., . ., .. …., …. ….** => WAIT

Steganography – Classical Examples

- Spy message
- Apparently neutral's protest is thoroughly discounted and ignored. Isman hard hit. Blockade issue affects pretext for embargo on byproducts, ejecting suets and vegetable oils
- Trick = Collect second letter from each word

Steganography – Classical Examples

- Apparently neutral's protest is thoroughly discounted and ignored. Isman hard hit. Blockade issue affects pretext for embargo on by products, ejecting suets and vegetable oils.

- pershingsailsfromnyjunei
- pershing sails from ny june i
- Pershing sails from NY June 1

Steganography – More Classical Examples

- **Character marking**
  - Selected letters of printed or typewritten text are over-written in pencil
  - The marks are ordinarily not visible unless the paper is held at an angle to bright light
- **Invisible ink**
  - A number of substances can be used for writing but leave no visible trace until heat or some chemical is applied to the paper

Steganography – More Classical Examples

- Pin punctures
  - Small pin punctures on selected letters are ordinarily not visible unless the paper is held up in front of a light
- Typewriter correction ribbon
  - Used between lines typed with a black ribbon, the results of typing with the correction tape are visible only under a strong light